



Danggui Buxue Tang improves therapeutic efficacy of doxorubicin in triple negative breast cancer via ferroptosis

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ABSTRACT

Ethnopharmacological relevance: Danggui Buxue Tang (DBT) has been used for over 800 years to enhance Qi and nourish Blood, and it is particularly beneficial for cancer patients. Recent research has shown that combining DBT with chemotherapy agents leads to superior anti-cancer effects, thereby enhancing therapeutic efficacy.

Aim of the study: The aim of this study was to evaluate the effectiveness of a combination therapy involving doxorubicin (DOX) and Danggui Buxue Tang (DBT) in the treatment of triple-negative breast cancer (TNBC) and to elucidate the underlying mechanisms of action.

Materials and methods: *In vitro* experiments were performed using MDA-MB-231 and 4T1 cells, while *in vivo* experiments were carried out using MDA-MB-231 xenograft mice. The therapeutic effects of the combination therapy were evaluated using various techniques, including MTT assay, colony formation assay, flow cytometry, transwell assay, immunofluorescence, transmission electron microscopy (TEM), histological analysis, western blotting, and bioluminescence assay.

Results: DBT was found to enhance DOX's anti-TNBC activity *in vitro* by promoting ferroptosis, as evidenced by the observed mitochondrial morphological changes using TEM. The combination therapy was also found to reduce the expression of Nrf2, HO-1, and GPX4, which are all targets for ferroptosis induction, while simultaneously increasing ROS production. Additionally, the combination therapy reduced nuclear accumulation and constitutive activation of Nrf2, which is a significant cause of chemotherapy resistance and promotes cancer growth. *In vivo* experiments using an MDA-MB-231 xenograft animal model revealed that the combination therapy significantly reduced tumor cell proliferation and accelerated TNBC deaths by modulating the Nrf2/HO-1/GPX4 axis, with no evidence of tissue abnormalities. Moreover, the combination therapy exhibited a liver protective effect, and administration of Fer-1 was able to reduce the ROS formation produced by the DBT + DOX combination therapy.

Conclusion: This study provides evidence that the combination therapy of DOX and DBT has the potential to treat TNBC by promoting ferroptosis through the Nrf2/HO-1/GPX4 axis.

1. Introduction

Triple-negative breast cancer (TNBC) is a subtype of breast cancer that lacks expression of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2). TNBC

accounts for approximately 15–20% of all breast cancer cases and is associated with aggressive tumor behavior and poor prognosis (Ganesan et al., 2021, 2023). Treatment options for TNBC are limited compared to other breast cancer subtypes, as targeted therapies like hormonal therapy or HER2-targeted agents are not effective (Wang et al., 2023b). Chemotherapy, particularly anthracyclines like doxorubicin (DOX), is

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Abbreviations

3-MA	3-methyladenine
AR	Astragali Radix
ASR	Angelicae Sinensis Radix
DBT	Danggui Buxue Tang
DOX	Doxorubicin
ER	Estrogen receptor
Fer-1	Ferrostatin-1
GPX4	Glutathione peroxidase 4
HER2	Human epidermal growth factor receptor 2
HO-1	Heme oxygenase 1
MDA	Malondialdehyde
Nrf2	Nuclear factor erythroid 2-related factor 2
PI	Propidium iodide
PR	Progesterone receptor
SOD	Superoxide dismutase
TNBC	Triple-negative breast cancer
Z-VAD	Z-VAD-FMK

commonly used in the treatment of TNBC. However, the efficacy of chemotherapy alone is often limited due to acquired resistance and toxic side effects (Zeng et al., 2023). Therefore, there is a need to explore combination therapies that can enhance the therapeutic impact of chemotherapy while minimizing adverse effects.

Danggui Buxue Tang (DBT) is a traditional Chinese herbal formula composed of Astragali Radix (Huangqi; AR) (roots of *Astragalus membranaceus* (Fisch.) Bunge or *Astragalus membranaceus* (Fisch.) Bunge var. *mongolicus* (Bunge) P.K. Hsiao) and Angelicae Sinensis Radix (Danggui; ASR) (root of *Angelica sinensis* (Oliv.) Diels) at the ratio of 5 : 1. DBT has been used for centuries in traditional Chinese medicine to tonify blood, boost the immune system, and improve overall health (Gong et al., 2016a, 2017; Kwan et al., 2019). The pharmacological properties of DBT have been recently investigated in both *in vivo* and *in vitro* studies (Gong et al., 2015, 2016c; Zheng et al., 2012). These studies have shown that the administration of DBT in various animal models results in several effects, such as the enhancement of red and white blood cell populations, stimulation of estrogenic properties, promotion of bone regeneration, triggering of immune responses, and induction of capillary and blood vessel formation (Gong et al., 2016b, 2018; Lin et al., 2017; Zhan et al., 2014; Zhang et al., 2013).

Ferroptosis is a form of regulated cell death characterized by the accumulation of iron-dependent lipid peroxides. It is distinct from other forms of cell death, such as apoptosis or necrosis (Lei et al., 2022; Zhang et al., 2021). Emerging evidence suggests that ferroptosis can be induced as a therapeutic strategy for cancer treatment, as cancer cells often exhibit increased sensitivity to ferroptosis compared to normal cells (Yang et al., 2023a). The nuclear factor erythroid 2-related factor 2 (Nrf2) pathway plays a crucial role in regulating cellular antioxidant responses and protecting cells from oxidative stress (Ganesan et al., 2020; Jayasuriya et al., 2021; V. Ganesh et al., 2022). However, aberrant activation of the Nrf2 pathway has been implicated in promoting cancer growth, metastasis, and chemotherapy resistance (Jayasuriya et al., 2022; Nagarajan et al., 2022). Nrf2 has been observed to be overexpressed in different types of tumors, including lung, breast, head and neck, uterine, ovary, and endometrium (Dodson et al., 2019; Feng et al., 2021; Fu et al., 2022; Shi et al., 2023; Si et al., 2020; Tang et al., 2022; Wang et al., 2022a; Yuan et al., 2021). In different types of tumors, Nrf2 and heme oxygenase 1 (HO-1) have been found to be frequently upregulated, and their elevated levels have been linked with tumor progression, aggressiveness, therapy resistance, and poor prognosis (Chang et al., 2018; Li et al., 2022b; Yang et al., 2021, 2024). HO-1, in particular, has a potent antioxidant and antiapoptotic effect that supports the

growth of cancer cells and increases their resistance to therapy (Yang et al., 2021). Activation of Nrf2 triggers the upregulation of HO-1 and glutathione peroxidase 4 (GPX4), which helps reduce oxidative stress and safeguard mitochondrial integrity, potentially preventing mitochondrial dysfunction (Yang et al., 2024). Excessive oxidative stress, commonly observed in different pathological conditions, can lead to ferroptosis, a unique type of cell death. Inhibiting the activation of antioxidant pathways, like Nrf2/HO-1, may result in increased oxidative stress, accumulation of intracellular iron, and the possible induction of ferroptosis (Cheng et al., 2023; Hassannia et al., 2019; Lei et al., 2022; Tang et al., 2022; Yang et al., 2014). Therefore, targeting the Nrf2/HO-1/GPX4 pathway has emerged as a potential therapeutic strategy for cancer treatment.

In recent years, research has demonstrated the anti-cancer properties of DBT and its ability to improve the effectiveness of chemotherapy in different types of cancer. However, whether DBT can enhance the therapeutic effects of chemotherapy specifically in treating TNBC is still unknown. Therefore, the aim of this study was to explore the potential of a combination therapy using DOX and DBT for treating TNBC, focusing on the disruption of the Nrf2/HO-1/GPX4 axis through ferroptosis induction.

2. Materials and methods

2.1. DBT formula preparation and the chromatogram identification

Chinese Pharmacopeia (2020) was utilized for the morphological identification of the AR and ASR. The voucher specimens were deposited at Zunyi Medical University in Zuhai, China (voucher # YP-201-07-01 for ASR and voucher # YP-201-07-02 for AR). AR and ASR were weighed in accordance with a 5:1 ratio and thoroughly mixed. The extraction process was repeated twice. The extracts were kept at -80°C after being lyophilized. The analysis was conducted applying an Agilent 1200 series system (Agilent, Waldbronn, Germany) that included a thermo-stated column compartment, an auto-sampler, a binary pump, and a degasser. Chromatographic separations were performed using acetonitrile (Solvent A) and 0.01% formic acid (Solvent B) in the mobile phase at a flow rate of 1.0 mL/min at room temperature on an Agilent, Eclipse Plus, C18 column (4.6×250 mm, $5\ \mu\text{m}$). Elution of the liner gradient was carried out with 15%–80% Solvent A for a duration of 0–70 min (Gong et al., 2023).

2.2. Cell culture

The 4T1, MDA-MB-231, and MCF-10A cell lines were obtained from the American Type Culture Collection (ATCC, Manassas, VA). These cell lines were cultured in Dulbecco's Modified Eagle's Medium with the addition of 100 g/L streptomycin, 100 IU/mL penicillin, and 10% fetal bovine serum. The cells were incubated at 37°C with 5% CO_2 in an incubator.

2.3. Cell viability assay

The viability of the cells was determined using the MTT test. MTT powder was provided by Sigma-Aldrich (St. Louis, MO, USA). Cells were plated and grown in 96-well plates. DOX doses ranged from 0.3 to $3\ \mu\text{M}$, and DBT herbal decoction concentrations ranged from 0.125 to $1.0\ \text{mg/mL}$. Working concentrations of Ferrostatin-1 (Fer-1), Z-VAD-FMK (Z-VAD), and 3-methyladenine (3-MA) were $1\ \mu\text{M}$, $10\ \text{nM}$, and $1\ \mu\text{M}$, respectively. All chemicals were purchased from Sigma-Aldrich. MTT solution at $0.5\ \text{mg/mL}$ was added to the cultures 48 h after treatment at $10\ \text{L/well}$. After 2 h of incubation, the purple crystal was dissolved in DMSO for the MTT experiment. Cell viability was calculated based on the percentage of absorbance in the control group, which was set to 100%.

2.4. Colony formation

One thousand cells were cultured in a 6-well plate and treated with different medicines for 48 h. After that, the medium was replaced with fresh medium and the cells were further incubated for 7 days. Following the incubation period, the cells were fixed in methanol at room temperature for 15 min. Subsequently, they were stained with crystal violet (Sigma-Aldrich) for 10 min (Cheng et al., 2023).

2.5. Flow cytometer

The cells were seeded and grown on 35-mm culture plates for 24 h prior to drug treatment. Annexin V-FITC/propidium iodide (PI) or SYTOX Fluorescence (Thermo Fisher Scientific, Grand Island, NY) were used to label the cells for flow cytometry. Before collection, the cells were washed twice with PBS and treated with 100 μ L of binding buffer for 15 min at room temperature in the dark. The samples were collected using a flow cytometry loader with a 10,000-event acquisition criterion for each tube. Quadrants were built based on the viable population, and the results were evaluated using the FlowJo v10.6 software.

2.6. Transwell

Briefly, 1×10^5 cells were grown in the upper transwell chambers in 200 μ L serum-free media and treated with different drugs. The lower chambers were filled with 600 μ L of 10% FBS medium. After 48 h, the non-migratory cells on the upper surface of the chamber were removed by scraping with a cotton swab, and the migrated cells were fixed in 4% paraformaldehyde. Subsequently, the migrated cells were stained with 0.2% crystal violet.

2.7. Transmission electron microscopy

After treatment with various drugs, cells were harvested using 0.25% trypsin with EDTA and washed with PBS. They were then pre-fixed with 2.5% glutaraldehyde soaked in 0.1 μ M sodium cacodylate buffer overnight at 4 °C. Thin sections, which were less than 100 nm thick, were cut with an ultramicrotome and stained with lead citrate and 2% uranyl acetate. The samples were imaged using a Philips CM100 transmission electron microscope (Hassannia et al., 2019).

2.8. Enzymatic activity

To measure intracellular SOD and MDA activities, Thermo Fisher Scientific's detection kit was used. After drug treatment, the cells were centrifuged and rinsed with PBS. The cells were then lysed using sample preparation solution, and the supernatants were collected for 5 min at 12,000 g at 4 °C. SOD activity was measured at 450 nm, and MDA was measured at 532 nm using a multimode microplate reader.

2.9. SDS-PAGE

The targeted protein expression in the cell lysate was determined by SDS-PAGE. After transferring the proteins to membranes, the membranes were incubated with the following antibodies: anti-GPX4 (CST, Danvers, MA), anti-SLC7A11 (CST), anti-Transferrin (CST), anti-NRF2, anti-HO-1 (CST), anti-tubulin (CST), all at 1:5000 dilutions, and anti-tubulin (CST), all at 1:5,000,000 dilutions. The immune complexes were identified using the enhanced chemiluminescence (ECL) technique (Amersham Biosciences, Piscataway, NJ) after a 3-h incubation period with secondary antibodies that had been HRP-conjugated. The band intensities were computed and examined using Image J.

2.10. Translocation

The translocation of NRF2 was identified using an Olympus Fluoview

FV1000 confocal system mounted on an Olympus microscope with a 63x objective. After the cells were fixed with 4% methanol-free paraformaldehyde treatment for 10 min, the antibody was diluted in cold PBS containing 2.5% fetal bovine serum and 0.1% Triton X-100 (Sigma-Aldrich). The samples were kept in a cold room overnight, and then a secondary antibody was used and incubated with a 1:1000 dilution of FITC-labeled anti-rabbit antibody for 3 h at room temperature in the dark. After incubation, the cells were washed twice, and DAPI (1:5000 dilution) was used to stain the nucleus.

2.11. Animals

Balb/c nude mice (8–10 weeks old) were obtained from Shanghai Laboratory Animal Company (Shanghai, China) and were housed in a pathogen-free environment with a 12-h light/dark cycle, a relative humidity of 70%, and a temperature of 22 ± 2 °C. MDA-MB-231 cells were subcutaneously implanted into the flanks of each mouse. Tumors were palpable and measurable six days after cell injection. The animals were randomly assigned to one of five treatment groups: the vehicle control group received Milli-Q water; the DOX group received DOX at a dose of 2 mg/kg; the DBT group received DBT herbal decoction at a dose of 10 mg/kg; the DBT + DOX group received 10 mg/kg of DBT and 2 mg/kg DOX; and the DBT + DOX + Fer-1 group received 10 mg/kg of DBT, 2 mg/kg DOX, and 2 mg/kg Fer-1. All experimental procedures adhered to the "Principles of Laboratory Animal Care" and were authorized by the Zunyi Medical University's Animal Experimentation Ethics Committee (No. 21-015 for Animal Ethics Approval). After receiving treatment for 21 days, the mice were euthanized, and the tumor weights were recorded.

2.12. Bioluminescence assay in the xenograft model

After being anesthetized with isoflurane, the mice received an intraperitoneal injection of 100 μ L of aminoluciferin (0.5 mM, diluted with normal saline). Three minutes later, bioluminescence imaging was performed using the PerkinElmer IVIS® Spectrum imaging system (Alameda, CA). The images were captured before the mice were euthanized for further research.

2.13. Immunohistochemical assay

The sections were immunostained using primary anti-Ki67 (CST) antibodies at a dilution of 1:200 and incubated with secondary antibodies at a dilution of 1:500. After staining, the slices were examined and photographed using a Nikon light microscope at a magnification of $200 \times$.

2.14. HE staining

Fresh heart, kidney, and liver tissues were fixed in 4% paraformaldehyde, dehydrated, embedded in paraffin, and then cut into 5 μ m thick slices. The slices were stained with hematoxylin for 30 min and eosin for 5 min. After staining, the slices were vitrified using xylene and sealed with neutral resin. The stained slices were analyzed and photographed using a Nikon light microscope at a magnification of $200 \times$.

2.15. Statistical analyses and other tests

To measure protein concentrations, BSA and the protein assay dye from Bio-Bradford Rad (Bio-Rad Laboratories, Hercules, CA) were used. Statistical analysis was performed using one-way ANOVA with Bonferroni's post-test.

3. Results

3.1. DBT enhances the cytotoxic effects of DOX

We have done the quality control of DBT prior to conducting the biological assessment, and the chromatography is published in Gong et al. (2023). In Fig. 1A, it was observed that DOX had an impact on the viability of TNBCs (4T1 and MDA-MB-231) as well as normal breast cells (MCF-10A). The results showed that DOX inhibited the proliferation of TNBCs in a dose-dependent manner, with 1 μ M being the optimal concentration for inducing cytotoxicity in cancer cells. Since 1 μ M of DOX did not affect cell viability in MCF-10A cells, it was chosen as the concentration for subsequent *in vitro* tests. Fig. 1B displays the assessment of cell viability of cancer and/or normal cells using the DBT herbal formula. The findings revealed that DBT had a dose-dependent negative impact on the survivability of TNBCs after 48 h of treatment. The effective concentration for reducing cell proliferation in both 4T1 and MDA-MB-231 was found to be 1 mg/mL. Notably, 1 mg/mL of DBT did not exhibit any cytotoxicity on MCF-10A cells, thereby indicating it as the optimal dose for this study. To further validate the cytotoxic efficacy and apoptotic status of the DBT and DOX combination therapy on TNBCs, colony formation and flow cytometry experiments were conducted (Fig. 1C&D). The results demonstrated that the combination group effectively restricted colony development compared to the single medicine group (Fig. 1C). Moreover, the Annexin V/PI dye results showed that the co-treatment group induced a higher rate of apoptosis compared to the other groups (Fig. 1D). These findings collectively suggested that DBT can enhance the cytotoxicity of DOX in TNBCs.

3.2. Combination therapy that inhibits cell proliferation is more closely related to ferroptosis

Fig. 2 demonstrates the use of pathway inhibitors to elucidate the mechanism of action of DBT cotreatment with DOX in suppressing TNBCs. Following 48 h of treatment, the ferroptosis inhibitor Fer-1 significantly reversed the inhibitory effects of the combination therapy on cell growth (Fig. 2A and B). However, the autophagy inhibitor 3-MA and the apoptosis inhibitor Z-VAD had no impact on restoring cell proliferation that had been reduced by the combination pharmaceutical therapy (Fig. 2A and B). Further investigation was done to examine the impact of Fer-1 in reversing the effects of DBT and DOX combination therapy on TNBC proliferation and migration using colony formation and cell migration experiments (Fig. 2C and D). The results revealed that the colony formation and migratory capacity of TNBC were significantly enhanced in the DBT + DOX + Fer-1 group as compared to the DBT + DOX group (Fig. 2C and D). However, DBT + DOX + Z-VAD or DBT + DOX + 3-MA did not improve colony formation or cell migration efficacies when compared to the DBT + DOX group (Fig. 2C&D). Overall, these findings suggested that DOX + DBT reduces cancer cell proliferation via ferroptosis.

Under electron microscopy, during ferroptosis, mitochondrial solidification, increased membrane density, and minimized or absent mitochondrial cristae were observed. In Fig. 3A, the mitochondrial envelopes of the control and DBT groups were intact, the mitochondrial crest was visible, and the form and size were regular. However, in the DBT + DOX group, the mitochondrial membrane density increased, the mitochondrial ridge reduced or vanished, and the mitochondrial architecture and volume were disordered (Fig. 3A). In comparison to the DBT + DOX group, the DBT + DOX + Fer-1 group had lower mitochondrial membrane density, a prominent mitochondrial crest, conventional mitochondrial architecture, and a somewhat higher volume (Fig. 3A). Considering that mitochondria are a major source of oxygen free radicals and that oxidative stress is linked to mitochondrial dysfunction, the level of reactive oxygen species (ROS) was evaluated in different treatment groups on 4T1 and/or MDA-MB-231 cells using a flow cytometer and a ROS detection kit (Fig. 3B). The results demonstrated that DOX

treatment increased ROS levels compared to the control group. However, the DBT + DOX group generated higher ROS content than the single drug group (Fig. 3B). In contrast, DBT + DOX + Fer-1 was able to lower the ROS content as reflected by the flow cytometer when compared to the DBT + DOX group (Fig. 3B).

3.3. Ferroptosis reduction impedes DBT-induced Nrf2/HO-1/GPX4 activation in DOX-treated cancer cells

Lipid peroxidation has been established as a crucial step in the process of ferroptosis (Chen et al., 2021; Shi et al., 2023). To investigate whether the combination therapy of DBT and DOX induces ferroptosis *in vivo*, we initially assessed the levels of malondialdehyde (MDA) and superoxide dismutase (SOD) in 4T1 and MDA-MB-231 cells (Fig. 4A and B). The results showed that MDA levels in the DBT + DOX combination therapy group were significantly higher than in the control group, while SOD activity was significantly lower (Fig. 4A and B). Importantly, the presence of Fer-1 altered the elevated MDA expression levels induced by the combination therapy (Fig. 4A and B). SLC7A11 and GPX4 are key targets for ferroptosis induction (Chen et al., 2023; Li et al., 2023; Yang et al., 2023b). Consequently, we examined the transcriptional levels of SLC7A11 and GPX4 after treatment with the combination therapy (Fig. 4C–F). The protein levels of SLC7A11 and GPX4 were significantly reduced in 4T1 and/or MDA-MB-231 cells following DBT + DOX treatment (Fig. 4C–F). Additionally, Fer-1 pretreatment reversed the expression levels of the proteins induced by the combination therapy (Fig. 4C–F). Furthermore, to further elucidate the induction of ferroptosis, we investigated the involvement of transferrin, a vital factor in iron accumulation during ferroptosis that facilitates iron import into cells (Khorsandi et al., 2020; Stockwell et al., 2017; Wang et al., 2022b; Yang et al., 2021). *In vitro*, the DOX + DBT combination therapy enhanced transferrin expression in 4T1 and/or MDA-MB-231 cells (Fig. 4C–F). However, Fer-1 was able to inhibit the enhanced translational activity of transferrin triggered by DOX + DBT after 48 h of treatment (Fig. 4C–F).

Under oxidative stress, nuclear accumulation and constitutive activation of Nrf2 increases, promoting cancer growth and serving as a major cause of chemotherapy resistance (Wang et al., 2023a; Yang et al., 2023b). Ferroptosis can be downregulated by the antioxidant system Nrf2/HO-1 through oxidative stress neutralization to enhance tumor sensitivity towards anticancer drugs (Li et al., 2022b; Yang et al., 2021, 2024). In this study, we conducted a western blotting test to measure the protein levels of Nrf2 and HO-1 (Fig. 5A–D). We found that DBT alone did not affect Nrf2/HO-1 expression, but it was significantly reduced after DBT + DOX treatment in TNBCs (Fig. 5A–D). Furthermore, the expression levels of proteins suppressed by DBT + DOX were reversed by Fer-1, as depicted in Fig. 5A–D. Additionally, we utilized laser confocal microscopy to investigate the activity of Nrf2 translocation in 4T1 and/or MDA-MB-231 cells (Fig. 5E). It was observed that neither DOX nor DBT alone had any impact on inhibiting Nrf2 translocation from the cytoplasm to the nucleus (Fig. 5E). In the control or single medication groups, Nrf2 fluorescence intensity was observed in both the cytosol and the nucleus (Fig. 5E). However, in the DBT + DOX group, the fluorescence was primarily localized in the cytoplasm, with minimal signal in the nucleus, as evidenced by co-localization with the DAPI signal (Fig. 5E). Nevertheless, pre-treatment with Fer-1 enhanced nuclear translocation activity, which had been inhibited in the DBT + DOX group (Fig. 5E). These findings suggest that the combination therapy of DBT + DOX can downregulate Nrf2/HO-1 expression and inhibit Nrf2 translocation. However, the addition of Fer-1 can reverse these effects, indicating its potential to modulate the Nrf2/HO-1 pathway in the context of ferroptosis and cancer treatment.

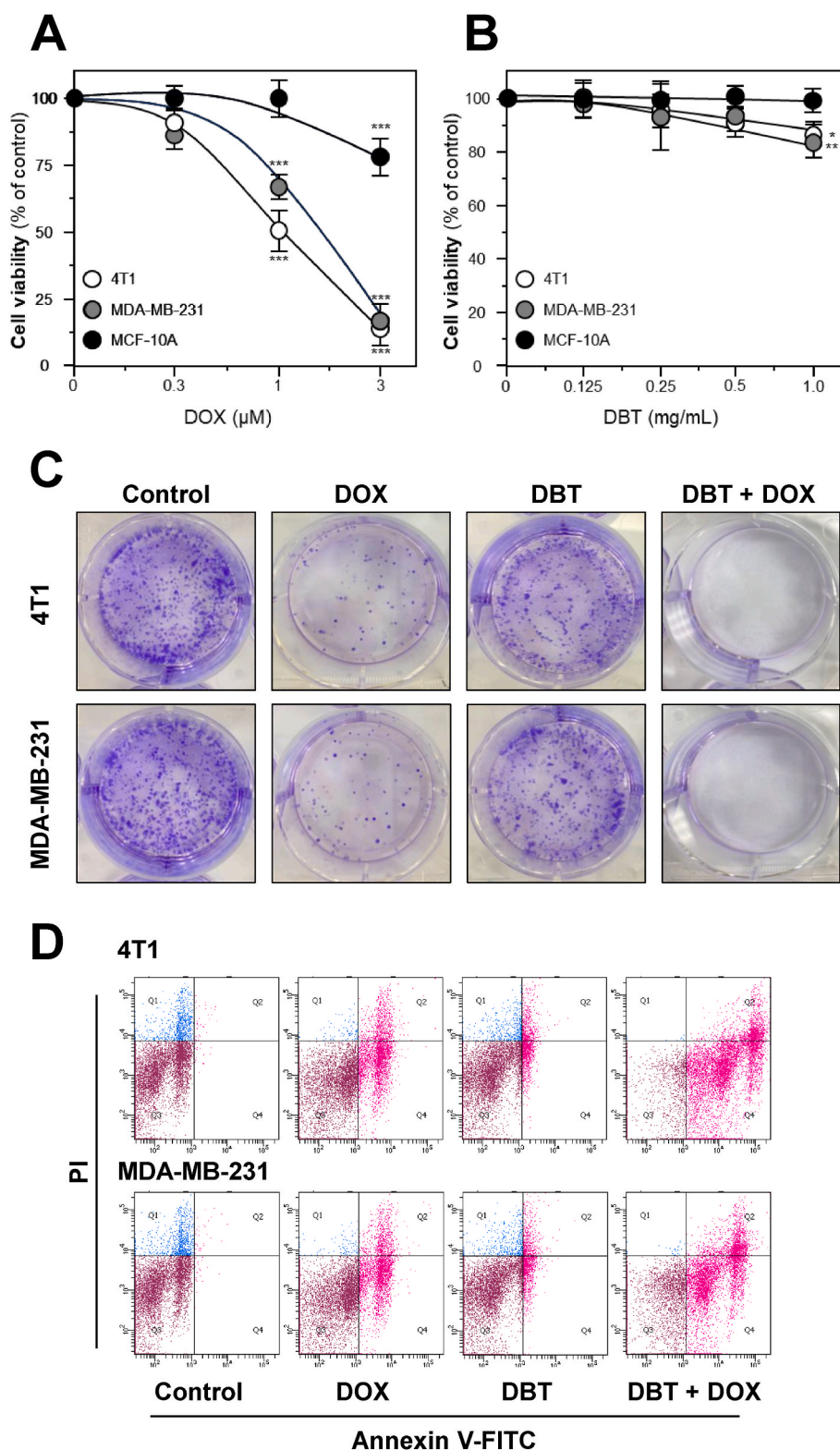


Fig. 1. DOX and DBT co-treatment mitigate cell viability.

[A] DOX (0.3–3 μM) and/or [B] (0.125–1.0 mg/mL) were applied to cultured cells for 48 h, and cell viability was determined using the MTT assay. [C] Cells were seeded in a 6-well plate after 48 h of drug treatment and switched with fresh medium for another 7 days. Crystal violet was used to stain the colony formation. DOX and DBT were used at concentrations of 1 μM and 1 mg/mL, respectively. [D] In the dual parametric dot plots combining annexin V-FITC and PI fluorescence, the viable cell population was shown in the bottom left quadrant, the early apoptotic cells in the bottom right quadrant, and the late apoptotic cells in the top right quadrant. Data are shown as a percentage change from the control group and in Mean $\pm$ SEM, with $n = 3$. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ was considered significant result as compared with control group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

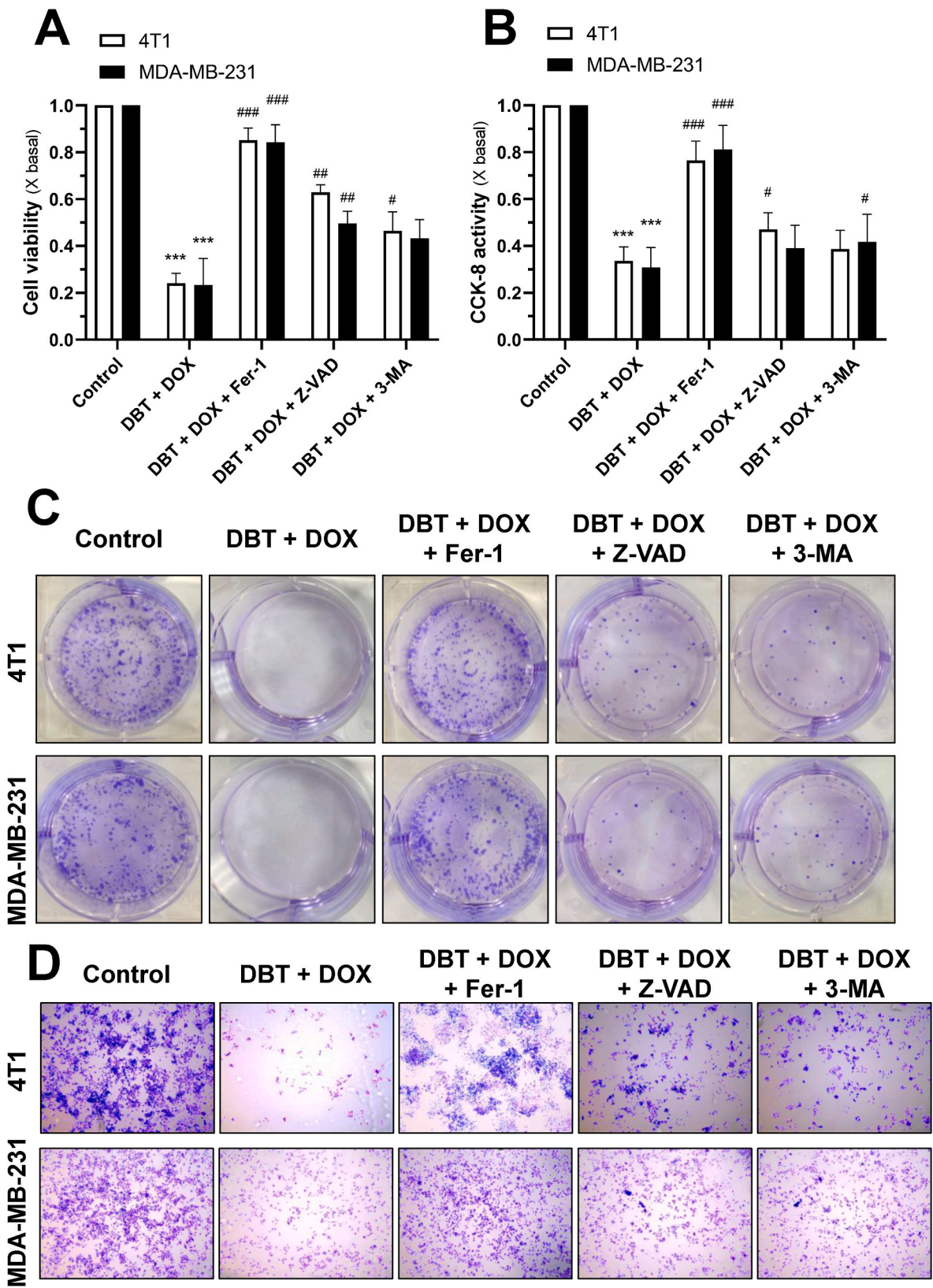


Fig. 2. DOX and DBT combination therapy induce cell death, which is reversed by Fer-1. [A-D] TNBC viabilities, CCK-8 activities, colony formation, and migration functions were determined following different drug treatment. TNBCs were pretreated for 3 h with Fer-1 (1 μ M), Z-VAD (10 nM), or 3-MA (1 μ M), and then incubated for 2 days with the combination therapy (1 μ M of DOX and 1 mg/mL of DBT). The data are presented as a fold change in comparison to the control group and in Mean $\pm$ SEM, with $n = 3$. *** $p < 0.001$ was considered a significant result when compared to the control group, while # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ were considered significant when compared to the combination therapy group.

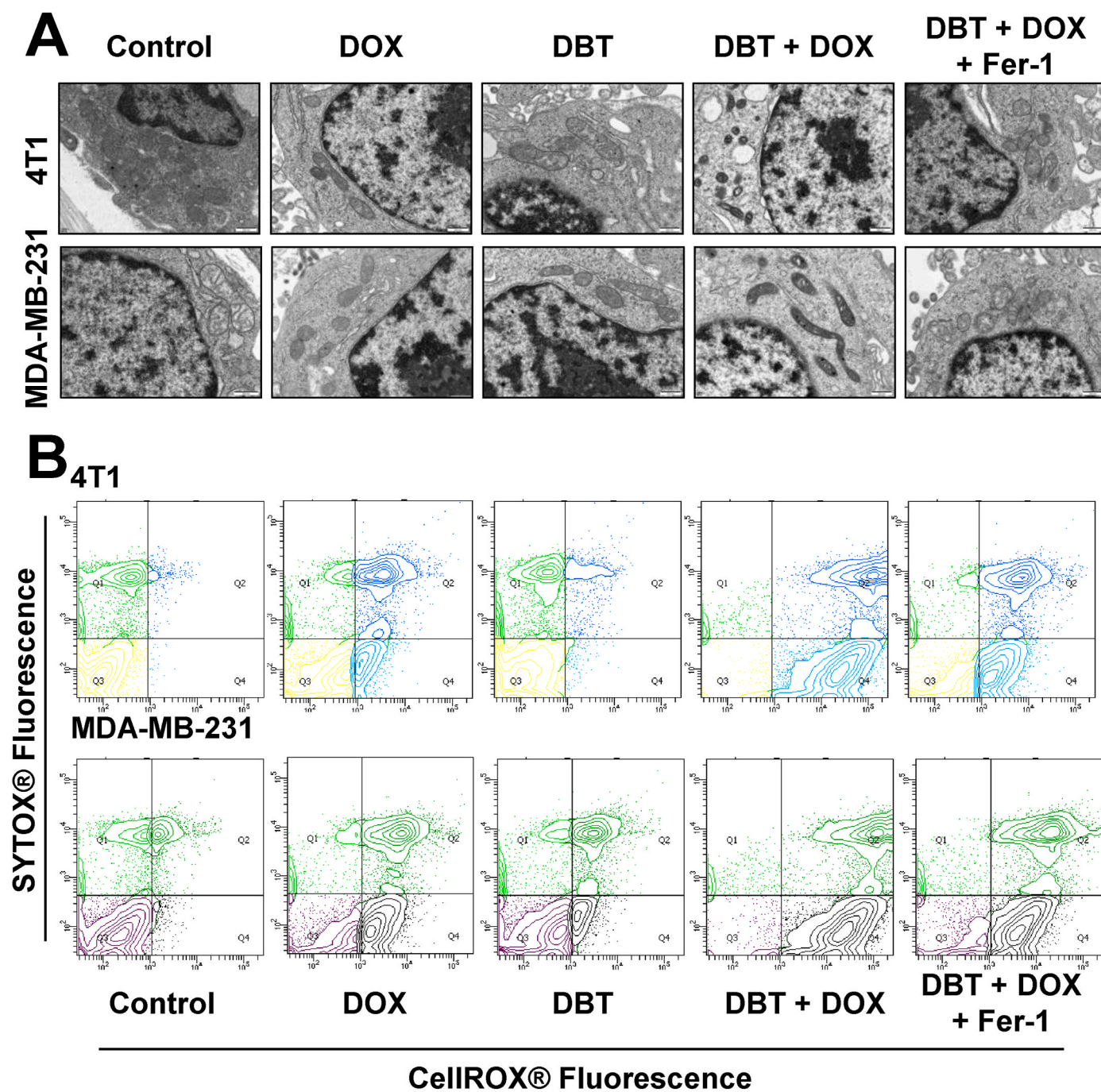


Fig. 3. Combination therapy causes alterations in the morphology of the mitochondria and increases the generation of ROS.

For 48 h, cultured cells were subjected to various drug treatments. **[A]** Flow cytometry was used to examine the fluorimetric with the CellROX® detection kit. **[B]** Combination therapy decreased mitochondria and crista in 4T1 and/or MDA-MB-231 cells, as detected by TEM. DOX, DBT, and Fer-1 were used at 1 μ M, 1 mg/mL, and 1 μ M concentrations, respectively.

3.4. DBT-mediated suppression of ferroptosis improves DOX's anti-cancer function *in vivo*

To assess if DBT could enhance the chemotherapeutic effect of DOX *in vivo*, a TNBC xenograft mouse model was established by injecting MDA-MB-231 cells into nude mice. After 21 days of treatment, the xenograft tumor size in the DOX + DBT group was significantly smaller compared to the single medication or control groups (Fig. 6A–C). However, when Fer-1 was administered in combination with DBT + DOX, the mean tumor volume was increased (Fig. 6A–C). To investigate the impact of DBT + DOX on cell proliferation and cell death, Ki67

immunohistochemistry and HE staining of tumor sections were performed (Fig. 6D). The combination therapy treatment significantly reduced tumor cell proliferation and increased TNBC cell death (Fig. 6D). HE staining was also conducted to examine the different organs isolated from the various treatment groups (Fig. 6E). Mice treated with DOX exhibited significant changes in the morphology of their livers and kidneys compared to the control group. However, no obvious tissue abnormalities were identified in the histological specimens isolated from the combination groups with or without Fer-1 treatment. Additionally, no significant evidence of heart fibrosis, liver inflammation, or glomerular or tubular abnormalities in the kidneys was detected

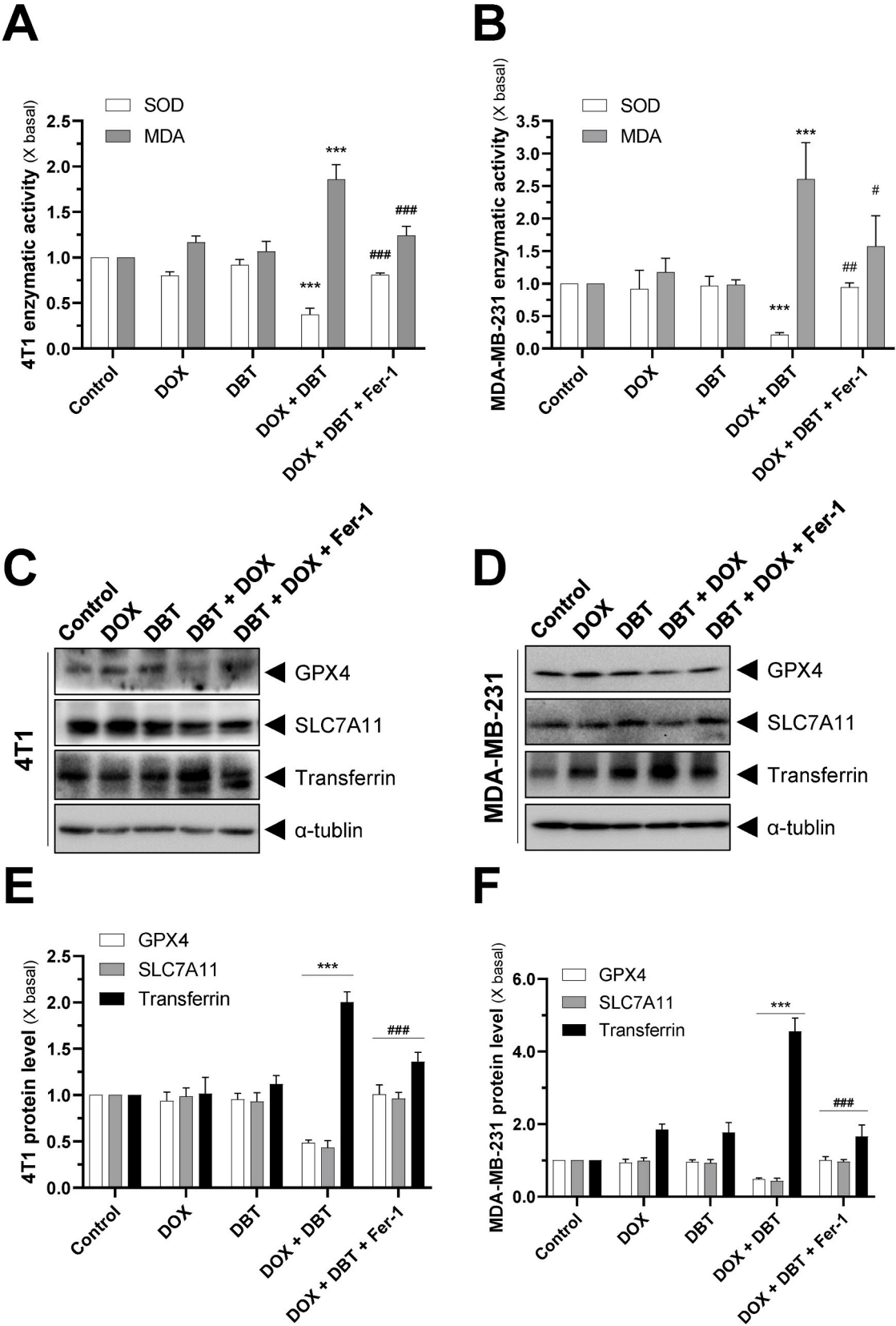


Fig. 4. Combination therapy causes changes in SOD and MDA via the GPX4/SLC7A11 axis. Cells were pretreated with or without Fer-1 for 3 h before being treated with various drugs for an additional 48 h [A&B] The kit was used to measure SOD and MDA enzymatic activity in 4T1 and/or MDA-MB-231 cells, respectively. [C–F] Western blots of the GPX4, SLC7A11, and Transferrin are shown here. When compared to the control group, $^{**}p < 0.01$ and $^{***}p < 0.001$ was considered a significant result, while $^{###}p < 0.001$ was considered significant when compared to the combination therapy group.

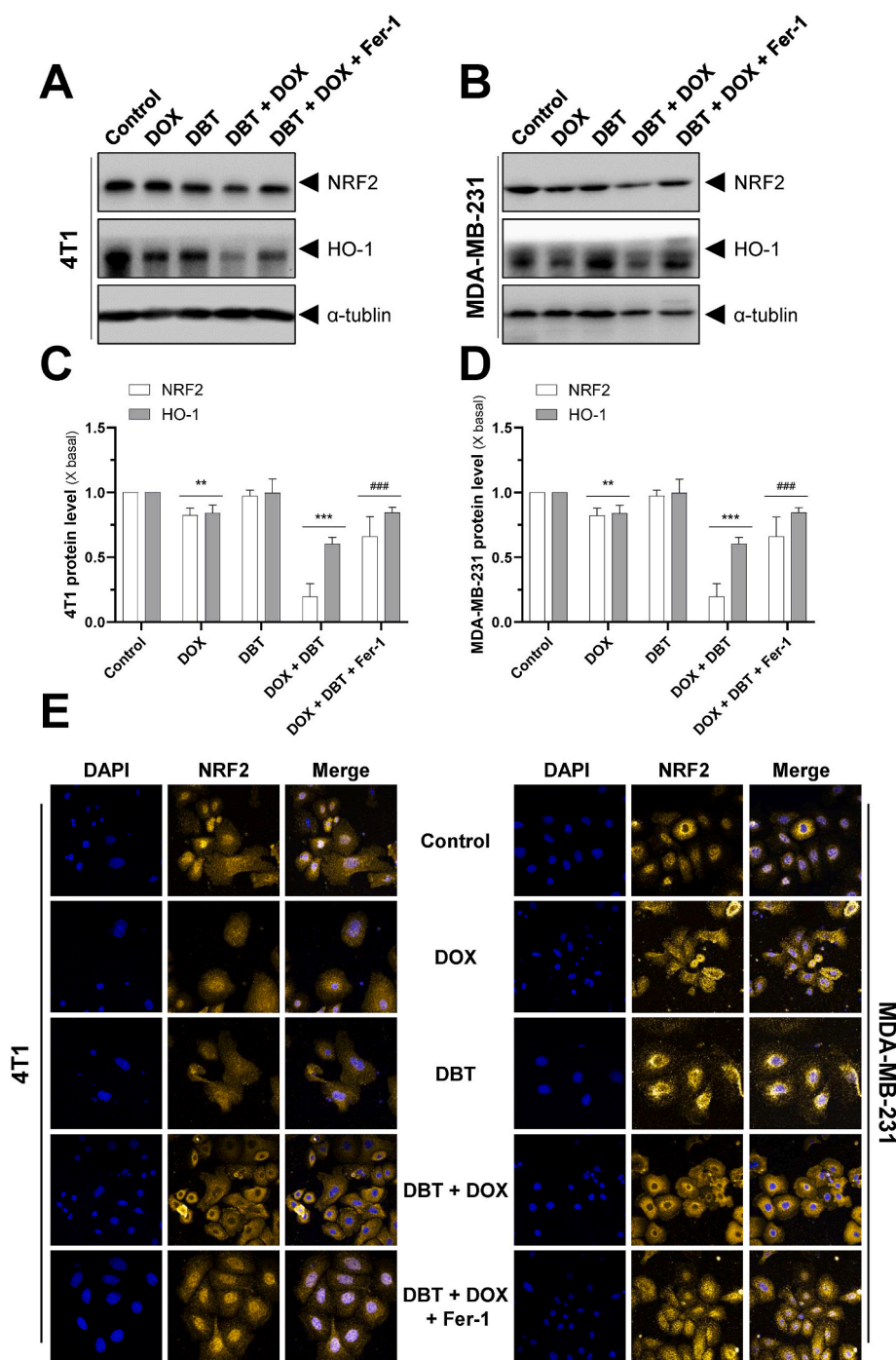


Fig. 5. Combination therapy reduces the translational levels of Nrf2 and HO-1.

Cells were pretreated for 3 h with or without Fer-1 before being treated for 48 h with various drugs. [A–D] Western blots of Nrf2 and HO-1 are shown here, with α -tubulin serving as a loading control. [E] The confocal microscope was used to examine the translocation functions of Nrf2 in cultured cells. DAPI (blue) was used to stain the nuclei, and the target protein was shown in yellow. ** $p < 0.01$ and *** $p < 0.001$ were considered significant when compared to the control group, while ### $p < 0.001$ was considered significant when compared to the combination therapy group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

(Fig. 6E). These findings are consistent with *in vitro* investigations that support the notion that DBT enhances DOX's anti-TNBC activity.

To further explore the effect of combination therapy with or without Fer-1 on liver damage, serum levels of ALT and AST were measured (Fig. 7A and B). The group treated with DOX demonstrated higher levels of ALT and AST. However, the increased levels of ALT and AST were significantly reduced in mice treated with DBT. Fer-1 treatment did not affect the liver-protective effect of DBT (Fig. 7A and B). Since ROS play a

crucial role in inducing ferroptosis, it was hypothesized that the combination therapy might lead to increased ROS production. Flow cytometry analysis was conducted to measure ROS expression levels, and the results confirmed that DBT + DOX significantly increased ROS production (Fig. 7C). Importantly, the administration of Fer-1 greatly reduced the formation of ROS induced by the DBT + DOX combination therapy (Fig. 7C). The mechanism by which DBT + DOX triggers ferroptosis *in vivo* was also investigated. Western blot analysis of tumor

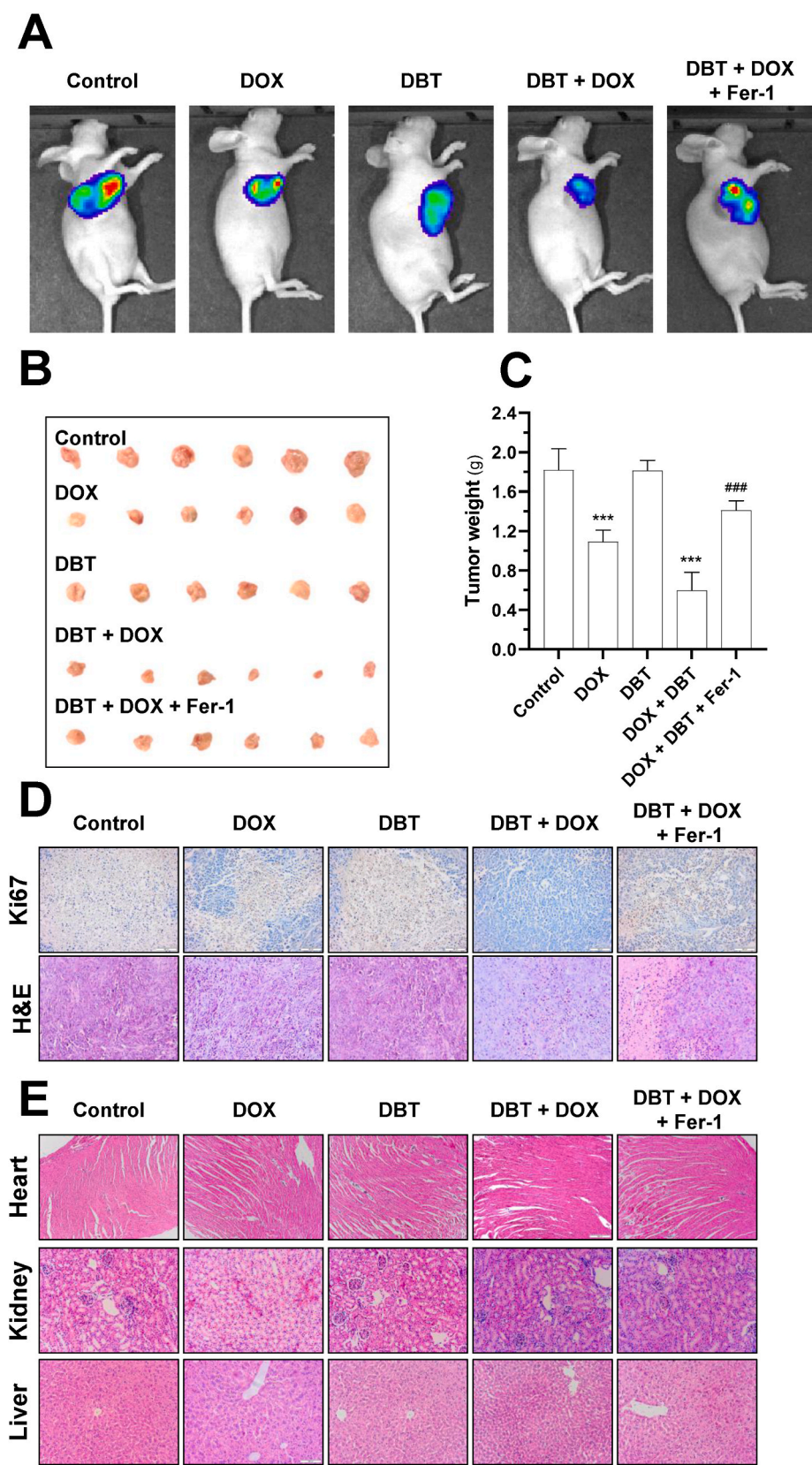


Fig. 6. Combination therapy demonstrates the anti-TNBCs' *in vivo* activities. The vehicle control group received Milli-Q water; the DOX group received 2 mg/kg DOX; the DBT group received 10 mg/kg DBT; the DBT + DOX group received 10 mg/kg DBT and 2 mg/kg DOX; and the DBT + DOX + Fer-1 group received 10 mg/kg DBT, 2 mg/kg DOX, and 2 mg/kg Fer-1. **[A]** Bioluminescence imaging of a nude mice xenograft tumor after 21 days of therapy. **[B]** Tumors pictures. **[C]** The tumor weight was determined. **[D]** Ki67 expression and histological results in tumor tissue. **[E]** HE staining of the heart, kidney, and liver in various groups.

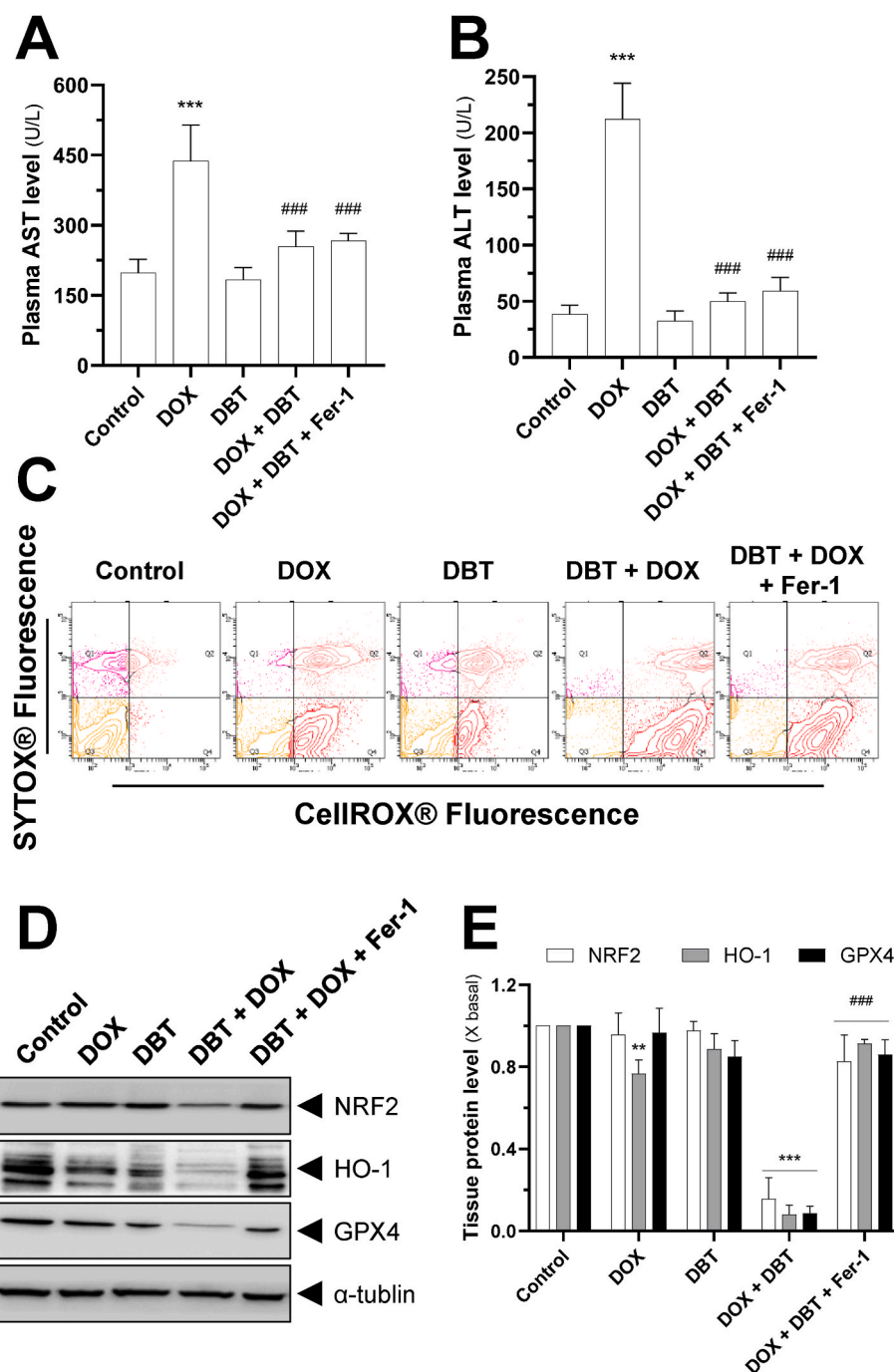


Fig. 7. Combination therapy has no effect on liver function, and the function of anti-TNBCs is linked to the downregulating Nrf2/HO-1/GPX4 axis.

[A&B] The kits used to measure AST and/or ALT levels in the xenograft nude mice were used. **[C]** Flow cytometry was used to determine ROS in homogenized tumor tissues. **[D]** Here were western blots results of Nrf2, HO-1, and GPX4 in homogenized tissues, with α-tubulin serving as a loading control. When compared to the control group, *** $p < 0.001$ was considered significant, while ### $p < 0.001$ was considered significant when compared to the combination therapy group.

tissue showed that DBT + DOX markedly reduced the expression of Nrf2, HO-1, and GPX4 (Fig. 7D&E). However, the administration of Fer-1 effectively restored the expression levels of Nrf2, HO-1, and GPX4 that were suppressed by the combination therapy (Fig. 7D&E). These findings strongly suggest that ferroptosis occurs both *in vitro* and *in vivo* during DOX + DBT-mediated tumor growth inhibition through the Nrf2/HO-1/GPX4 axis.

4. Discussion

Breast cancer is a complex and diverse disease, and TNBC is a

particularly challenging subtype due to the lack of targeted treatment options available. Chemotherapy, such as DOX, is commonly used in TNBC treatment, but its efficacy is often limited by acquired resistance and adverse effects (Akbaribazm et al., 2020; Kazantseva et al., 2022; Tang et al., 2022). Therefore, it is necessary to explore novel therapeutic approaches that can enhance the effectiveness of chemotherapy while minimizing toxicity. In this study, we investigated the potential of a combination therapy involving DOX and DBT in treating TNBC. Our hypothesis was that this combination therapy could promote ferroptosis, a regulated form of cell death characterized by the accumulation of iron-dependent lipid peroxides.

The two herbs, AR and ASR, which are present in DBT, have a long history of use in Traditional Chinese Medicine and are well-known for their diverse pharmacological properties in Asian countries (Gong et al., 2017, 2023). These herbs possess a wide range of pharmacological effects, including anti-inflammatory, antioxidant, and anticarcinogenic properties (Kwan et al., 2019; Lin et al., 2017; Zhan et al., 2014; Zhang et al., 2013). In a previous study, the effects of DBT during menopause on breast cancer cell growth were evaluated. The study demonstrated that the ethanolic extract of the dried root of ASR decreased the growth of MCF-7 cells *in vitro* and in MCF-7 xenograft mice models (Zhu et al., 2021). Similarly, another study indicated that the aqueous extract of ASR abolished the growth of MCF-7 cells, possibly through a weak estrogen-agonistic activity (Wang et al., 2020; Zhan et al., 2014). Ferulic acid, the active compound found in ASR, has been reported to inhibit the proliferation of breast cancer cells by down-regulating the expression of HER2 (Chang et al., 2006). Furthermore, combined treatment of DBT with a chemotherapeutic drug has been shown to alleviate cancer symptoms and improve the quality of life in patients (Bamodu et al., 2019). However, there have been no reports specifically investigating the effects of ASR in TNBC using *in vitro* and *in vivo* models.

Inhibiting tumor cell proliferation is a crucial strategy for cancer treatment (Ganesan and Xu, 2017). Therefore, we investigated the impact of DBT on the growth of TNBC cells, MDA-MB-231 and 4T1, using MTT and colony formation assays. The results demonstrated that DBT, either alone or in combination with DOX, inhibited TNBC cell proliferation. Additionally, flow cytometry analysis revealed that treatment with DBT, either alone or in combination with DOX, significantly increased apoptotic rates. Furthermore, co-administration of DBT further enhanced cell death and ROS generation in TNBC cells. Increased ROS production has been associated with anti-tumorigenic effects, as it can trigger cell death pathways. Our findings suggest that DBT may exert its anti-tumor effects through the generation of ROS, contributing to the inhibition of TNBC cell growth. These findings are consistent with a previous study that showed DBT synergistically induced apoptosis and autophagosomes in spermatocytes (Zhang et al., 2024). It is worth noting that ROS production can also contribute to anti-tumorigenic effects by initiating tumor cell death through oxidative stress (Li et al., 2022a; Zhang et al., 2022b).

Ferroptosis has gained increasing attention as a potential therapeutic strategy for cancer treatment, as cancer cells often exhibit increased sensitivity to ferroptosis compared to normal cells (Zhang et al., 2022a). In our *in vitro* experiments, we demonstrated that DBT enhanced the anti-TNBC activity of DOX by promoting ferroptosis, resulting in a significant reduction in tumor cell proliferation. The combination treatment involving an autophagy inhibitor and apoptosis inhibitor presents a promising opportunity for the discovery and development of novel and highly effective therapeutic approaches in TNBC treatment (Liu et al., 2020; Lu et al., 2023). In this study, we investigated the impact of treating TNBC cells with an autophagy inhibitor or apoptosis inhibitor in addition to the combination therapy of DBT and a chemotherapeutic drug. Our results showed that there were no significant changes in colony formation or cell migration efficacy when compared to the DBT + DOX combination therapy group. These outcomes were consistent with earlier studies (Lei et al., 2021; Zhang et al., 2022a). Therefore, our results suggest that the combination therapy of DBT and DOX may be sufficient for promoting ferroptosis and inducing cell death in TNBC cells, and the addition of autophagy or apoptosis inhibitors may not provide further benefits.

Our study revealed that the combination of DBT and DOX exhibited a synergistic effect on TNBC cells, leading to enhanced cell death and reduced cell proliferation compared to DOX treatment alone. Notably, we observed that the combination treatment induced ferroptosis, a recently discovered form of programmed cell death that has been implicated in cancer cell death. Ferroptosis involves the accumulation of iron-dependent lipid peroxides, upregulation of transferrin, generation of MDA, and decreased activation of SOD and GPX4. Our findings are

consistent with previous studies that have shown the promotion of ferroptosis and generation of ROS by the combination therapy (Fu et al., 2022; Yuan et al., 2021). These findings highlight the potential of the DBT and DOX combination therapy in exploiting the ferroptosis pathway for TNBC treatment. By targeting ferroptosis, this combination therapy may provide a promising therapeutic strategy for TNBC patients, who currently have limited treatment options.

Importantly, the combination therapy downregulated the expression of Nrf2, HO-1, and GPX4, which are key regulators of ferroptosis induction. This suggests that the combination therapy-induced ferroptosis is mediated through the Nrf2/HO-1/GPX4 axis. Furthermore, the combination therapy was found to inhibit the nuclear accumulation and constitutive activation of Nrf2, a transcription factor that plays a crucial role in cellular antioxidant responses. Aberrant activation of the Nrf2 pathway has been implicated in promoting cancer growth, metastasis, and chemotherapy resistance (Feng et al., 2021; Shi et al., 2023; Wang et al., 2022a; Yuan et al., 2021). In the present study, we observed a significant inhibition of Nrf2 activation by DBT, which in turn promoted the anti-cancer activity of DOX. The combination therapy also showed a liver protective effect, and the administration of Fer-1, a ferroptosis inhibitor, reduced the production of ROS induced by the combination therapy. These findings suggest that the Nrf2/HO-1/GPX4 axis plays a critical role in the ferroptosis induction and anti-cancer effects of the combination therapy. Targeting this pathway may offer a novel therapeutic approach for TNBC treatment and potentially overcome chemotherapy resistance.

In the *in vivo* experiments using a TNBC xenograft mouse model, the combination therapy of DOX and DBT significantly reduced tumor size, tumor weight, and accelerated TNBC deaths. These findings provide strong evidence for the potential of the combination therapy as an effective treatment strategy for TNBC. Importantly, the combination therapy also showed a liver protective effect, as evidenced by the reduction in serum ALT and AST levels. This is a significant finding, as liver toxicity is a common side effect of chemotherapy (Ganesan et al., 2017; Sukalingam et al., 2018). To further understand the mechanism of action, we investigated the role of ROS in the combination therapy-induced ferroptosis. The results showed that the combination therapy increased ROS production, confirming the hypothesis that the therapy triggers the production of ROS. However, the administration of Fer-1, a ferroptosis inhibitor, significantly reduced the ROS formation induced by the combination therapy. This suggests that targeting ROS and ferroptosis may be a viable approach to enhance the therapeutic efficacy of the combination therapy. These outcomes were consistent with earlier studies that have also highlighted the role of ROS and ferroptosis in the anti-cancer effects of the combination therapy (Lei et al., 2021; Zhang et al., 2022a). Together, these findings provide valuable insights into the underlying mechanisms of the combination therapy and support its potential as a promising therapeutic strategy for TNBC.

Overall, the results of this study provide valuable insights into the potential of the combination therapy involving DOX and DBT for the treatment of TNBC. The combination therapy promotes ferroptosis, inhibits the Nrf2 pathway, and demonstrates a liver protective effect, suggesting that it may offer a promising therapeutic approach for TNBC treatment. These findings highlight the importance of considering alternative treatment strategies that can enhance the effectiveness of chemotherapy while minimizing toxicity. Further research and clinical trials are warranted to validate these findings and explore the translational potential of the combination therapy in TNBC patients. The identification of the Nrf2/HO-1/GPX4 axis as a key mediator of ferroptosis induction and anti-cancer activity provides a potential target for future drug development. Moreover, the liver protective effect observed in this study suggests that the combination therapy may have a favorable safety profile.

5. Conclusion

This study provides evidence for the potential of the combination therapy of DOX and DBT in the treatment of TNBC. The combination therapy enhances the anti-TNBC activity of DOX through the promotion of ferroptosis and inhibition of the Nrf2 pathway. It leads to a reduction in tumor cell proliferation and demonstrates a liver protective effect. These findings suggest that the combination therapy may serve as an effective treatment strategy for TNBC. However, further research and clinical trials are necessary to validate the efficacy and safety of the DOX and DBT combination therapy in TNBC patients.

Ethics approval and consent to participate

All experiments conducted in this study were performed in compliance with the Guidelines for Laboratory Animal Care and approved by the Committee on the Use of Live Animals in Teaching and Research.

CRediT authorship contribution statement

Guowei Gong: Writing – original draft, Validation, Investigation, Funding acquisition, Conceptualization. **Kumar Ganesan:** Writing – review & editing. **Yaqu Liu:** Formal analysis. **Yongping Huang:** Formal analysis. **Yuting Luo:** Formal analysis. **XueXu Wang:** Formal analysis. **Zhenxia Zhang:** Methodology, Formal analysis. **Yuzhong Zheng:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.117655>.

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